

3/23/2023 Pin-jointed Pre-Lab

Design the lab by myself!

## Introduction

The aims of the pin-jointed frame lab are:

- to observe and compare the behaviour of statically-determinate and statically-indeterminate pin-jointed frameworks when applying force on it;
- to compare the experimentally observed behaviour with predictions from theory.

## Apparatus

- CST114
- CST100 Structures Platform
- Temperature =  $+5^{\circ}\text{C}$  to  $+40^{\circ}\text{C}$
- Humidity 湿度 Range =  $80\%$  ( $T < 31^{\circ}\text{C}$ )  $\xrightarrow{\text{linearly}}$   $50\%$  ( $40^{\circ}\text{C}$ )
- Hexagon Tools
- Inclinator 倾角计
- Supports: Pinned  $\odot$  ( $2.8\text{kg}$ ) and Roller  $\uparrow\odot$  ( $2.7\text{kg}$ )
- Upright: Hold supports ( $3\text{kg}$ )
- Load cell: Max capacity  $650\text{N}$  ( $1.85\text{kg}$ )

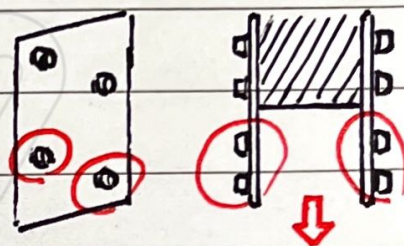


3/23/2023 Pin-jointed Pre-Lab

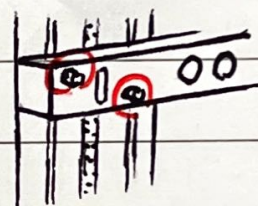
## Procedure

### 1. Setup

(i) Fixing Upright to CST 100 Structure Platform by M8 Fixings (lower 2 fixings at each side)



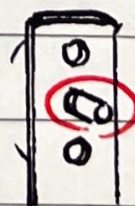
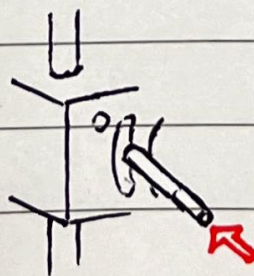
(ii) Fitting the Supports by M8 Fixings (2 fixing at each side at each support - 2 supports in total)



(iii) Using the Inclinator to make sure supports are tangent to the Upright by reading

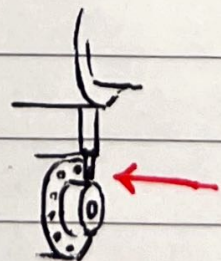
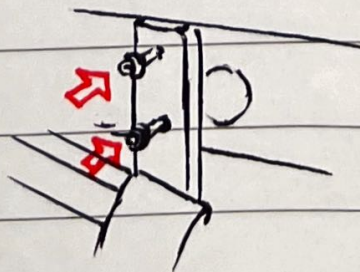


(iv) Fitting the Load Cell Locking Pin, and using Inclinator to make sure the struss is horizontal for the lowest one



(v) Placed Support Pin in Middle hole of support

(vi) Fitting the Trammel and Dial Indicator by M4 Fixings on the upper support





3/23/2023 Pin-jointed Pre-Lab

- (vii) Fitting the Strain Gauge Amplifier to one of the rear beams of the platform and use the cable supplied to connect it's DIGITAL OUTPUT socket to any DIGITAL INPUT sockets on the HUB of the CSTno Platform
- (viii) Connecting the load cell to any of the DIGITAL INPUT sockets on the HUB of the CSTno Platform
- (ix) Fitting the battery into the Dial Indicator
- (x) Connecting the cable to the Interface Hub

## 2. Statically ~~Indeterminate~~ Determine

- i) Checking the temperature and humidity of the laboratory and record if possible. the experiment could not start when temperature and humidity out of range
- ii) Checking if all cables are connected correctly and all devices is turn on
- iii) Setting the Dial Indicator to zero
- iv) Input Calibration Factor of cables and input them into software
- v) Select 'Determinate' tab in software
- vi) Apply load 25N each time from 0N to 400N on load-cell. record deflection and click the record data



3/23/2023 Pin-jointed Pre-Lab

(vii) Reduce load 20N each time from 400N to 0N  
same as (vi)

\* Noticed that pre-load is 200N

### 3. Statically Indeterminate

same as process above, despite change tab to 'Indeterminate' at step (v) and add truss '8'

### 4. After experiment

i) Reset all reading to zero

ii) Deallocate CST114 and CST100, go back to the original states

### 5. Preliminary Data Processing

i) Plot the deflection curve (Load vs. Deflection) with the experiment value compare to the prediction value to verify the experiment

### 6. Further Data Processing

i) Plot the force readings for each truss with load, compare them with the prediction from theory

ii) What could possibly go wrong?

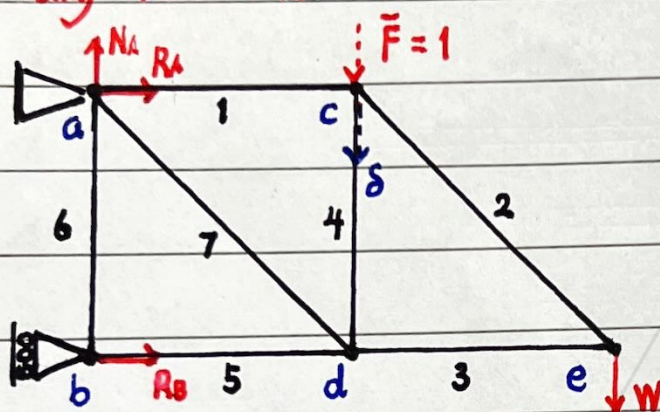
iii) Plot the error bar by the recorded error from lab alongside the curve.



3/23/2023 Pin-jointed Pre-Lab

Prediction by theory

Statically Determine

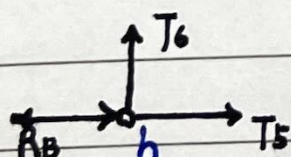


$$M = b + r - 2j = 0$$

$$\Sigma F_x (\rightarrow) = N_A - W = 0 \quad \therefore N_A = W$$

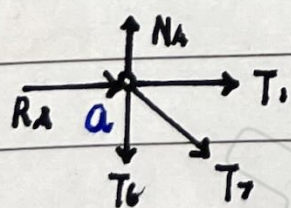
$$\Sigma F_y (\uparrow) = R_A + R_B = 0 \quad \therefore R_A = -R_B$$

$$M_b (\circlearrowleft) = R_A L + W(2L) = 0 \quad \therefore R_A = -2W \quad \therefore R_B = 2W$$



$$(\rightarrow) R_B + T_5 = 0 \quad \therefore T_5 = -2W$$

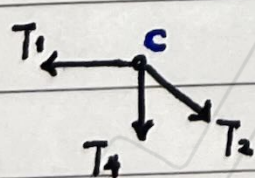
$$(\uparrow) T_6 = 0$$



$$(\rightarrow) R_A + T_1 + \frac{\sqrt{2}}{2} T_7 = 0$$

$$(\uparrow) T_6 + \frac{\sqrt{2}}{2} T_7 = N_A \quad \therefore T_7 = \sqrt{2}W$$

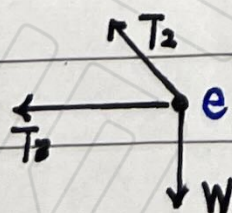
$$\therefore T_1 = W$$



$$(\rightarrow) T_1 = \frac{\sqrt{2}}{2} T_2 \quad \therefore T_2 = \sqrt{2}W$$

$$(\uparrow) T_4 + \frac{\sqrt{2}}{2} T_2 = 0 \quad \therefore T_4 = -W$$

$$(\rightarrow) T_3 + \frac{\sqrt{2}}{2} T_2 = 0 \quad \therefore T_3 = -W$$





3/23/2023 Pin-jointed Pre-Lab

For virtual work,  $\bar{F} = 1$

$$\therefore N_A = 1, \bar{R}_A = -\bar{R}_B = -1$$

$$\text{At } b, \bar{T}_5 = -1, \bar{T}_6 = 0$$

$$\text{At } a, \bar{T}_7 = \sqrt{2}, \bar{T}_1 = 0$$

$$\text{At } c, \bar{T}_2 = 0, \bar{T}_4 = -1$$

$$\text{At } e, \bar{T}_3 = 0$$

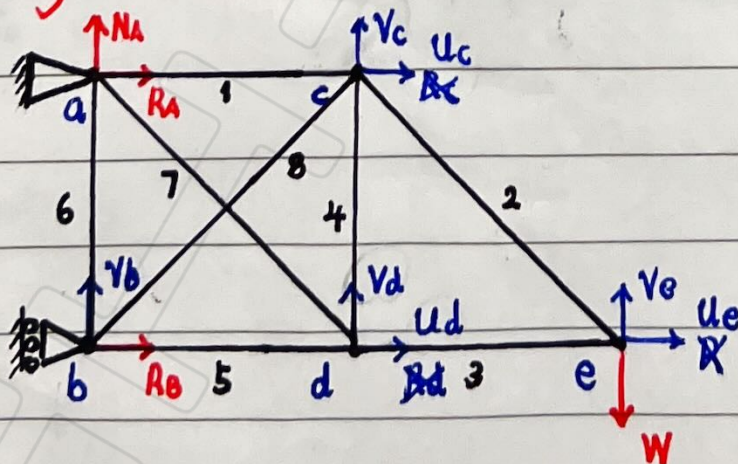
$$\begin{aligned} \therefore 1 \cdot \delta &= \bar{T}_1 e_1 + \bar{T}_2 e_2 + \dots + \bar{T}_7 e_7 \\ &= 0 + 0 + 0 + (-1) \cdot \left( \frac{-WL}{AE} \right) + (-1) \cdot \left( \frac{-2WL}{AE} \right) \\ &\quad + 0 + \sqrt{2} \cdot \frac{(\sqrt{2}W)(\sqrt{2}L)}{AE} \end{aligned}$$

$$\therefore \delta (\downarrow) = (1 + 2 + 2) \frac{WL}{AE}$$

$$\therefore T_1 = W, \quad T_2 = \sqrt{2}W, \quad T_3 = -W, \quad T_4 = -W,$$

$$T_5 = -2W, \quad T_6 = 0, \quad T_7 = \sqrt{2}W$$

Statically Indeterminate



$$M = b + r - 2j = 1$$



3/23/2023 Pin-jointed Pre-Lab

$$e_1 = u_o = T_1 \frac{L}{AE}$$

$$\therefore T_1 = u_c \frac{AE}{L}$$

$$e_2 = \frac{\sqrt{2}}{2} (v_c - v_e + u_e - u_c) = T_2 \frac{\sqrt{2}L}{AE}$$

$$\therefore T_2 = \frac{v_c - u_c - v_e + u_e}{2} \frac{AE}{L}$$

$$e_3 = u_c - u_d = T_3 \frac{L}{AE}$$

$$\therefore T_3 = (u_e - u_d) \frac{AE}{L}$$

$$e_4 = v_c - v_d = T_4 \frac{L}{AE}$$

$$\therefore T_4 = (v_c - v_d) \frac{AE}{L}$$

$$e_5 = u_d = T_5 \frac{L}{AE}$$

$$\therefore T_5 = u_d \frac{AE}{L}$$

$$e_6 = -v_b = -T_6 \frac{L}{AE}$$

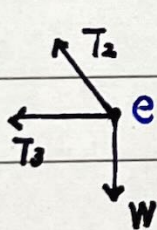
$$\therefore T_6 = -v_b \frac{AE}{L}$$

$$e_7 = \frac{\sqrt{2}}{2} (v_c - v_d + u_d) = T_7 \frac{\sqrt{2}L}{AE}$$

$$\therefore T_7 = \frac{u_d - v_d}{2} \frac{AE}{L}$$

$$e_8 = \frac{\sqrt{2}}{2} (v_c - v_b + u_c) = T_8 \frac{\sqrt{2}L}{AE}$$

$$\therefore T_8 = \frac{v_c - v_b + u_c}{2} \frac{AE}{L}$$

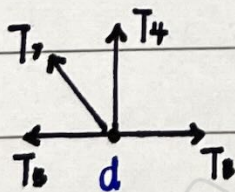


$$(\uparrow) \frac{\sqrt{2}}{2} T_2 = W$$

$$\therefore \frac{\sqrt{2}}{4} (v_c - u_c - v_e + u_e) \frac{AE}{L} = W \quad (1)$$

$$(\rightarrow) T_3 + \frac{\sqrt{2}}{2} T_2 = 0$$

$$\therefore (u_e - u_d) \frac{AE}{L} = -W \quad (2)$$

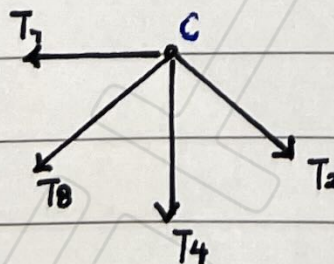


$$(\uparrow) T_4 + \frac{\sqrt{2}}{2} T_7 = 0$$

$$\therefore (v_c - v_d - \frac{\sqrt{2}}{4} v_d + \frac{\sqrt{2}}{4} u_d) \frac{AE}{L} = 0 \quad (3)$$

$$(\rightarrow) T_3 = T_6 + \frac{\sqrt{2}}{2} T_7$$

$$\therefore (u_d + \frac{\sqrt{2}}{4} u_d - \frac{\sqrt{2}}{4} v_d) \frac{AE}{L} = -W \quad (4)$$



$$(\uparrow) T_4 + \frac{\sqrt{2}}{2} T_8 + \frac{\sqrt{2}}{2} T_2 = 0$$

$$+ \frac{\sqrt{2}}{4} u_e - \frac{\sqrt{2}}{4} v_e = 0$$

$$\therefore (v_c + \frac{\sqrt{2}}{4} v_c + \frac{\sqrt{2}}{4} v_c - v_d - \frac{\sqrt{2}}{4} v_b + \frac{\sqrt{2}}{4} u_c - \frac{\sqrt{2}}{4} u_c) \frac{AE}{L} = 0 \quad (5)$$

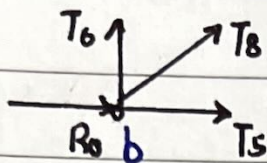
$$(\rightarrow) T_1 + \frac{\sqrt{2}}{2} T_8 = \frac{\sqrt{2}}{2} T_2$$

$$\therefore (u_c + \frac{\sqrt{2}}{4} u_c + \frac{\sqrt{2}}{4} v_c - \frac{\sqrt{2}}{4} v_b) \frac{AE}{L} = W \quad (6)$$

$$* (5) \rightarrow (v_c + \frac{\sqrt{2}}{4} v_c - v_d - \frac{\sqrt{2}}{4} v_b + \frac{\sqrt{2}}{4} u_c) \frac{AE}{L} = -W$$



3/23/2023 Pin-jointed Pre-Lab



$$(\uparrow) T_6 + \frac{\sqrt{2}}{2} T_8 = 0$$

$$\therefore -V_b - \frac{\sqrt{2}}{4} V_b + \frac{\sqrt{2}}{4} V_0 + \frac{\sqrt{2}}{4} U_c = 0$$

(7)

From (2),  $U_e = U_d - \frac{WL}{AE}$

to (4),  $V_d = 2\sqrt{2} \frac{WL}{AE} + (U_d + \frac{\sqrt{2}}{4} U_d) \frac{4}{\sqrt{2}} = (1+2\sqrt{2}) U_d + 2\sqrt{2} \frac{WL}{AE}$

to (3),  $V_c = (1 + \frac{\sqrt{2}}{4}) V_d - \frac{\sqrt{2}}{4} U_d = (2+2\sqrt{2}) U_d + (1+2\sqrt{2}) \frac{WL}{AE}$

to (5) and (6),  $U_c = 2 \frac{WL}{AE} + V_c - V_d = U_d + 3 \frac{WL}{AE}$

$$V_b = (1+2\sqrt{2}) U_c + V_c - 2\sqrt{2} \frac{WL}{AE}$$

$$= (3+4\sqrt{2}) U_d + (4+6\sqrt{2}) \frac{WL}{AE}$$

to (1),  $-(5 + \frac{14}{4}\sqrt{2}) U_d - (1+7\sqrt{2}) \frac{WL}{AE} + (1 + \frac{\sqrt{2}}{2}) U_d + (1 + \frac{\sqrt{2}}{4}) \frac{WL}{AE} + \frac{\sqrt{2}}{4} U_d + \frac{3}{4}\sqrt{2} \frac{WL}{AE} = 0$

$$\therefore -(4+4\sqrt{2}) U_d = (6+6\sqrt{2}) \frac{WL}{AE} \quad \therefore U_d = -\frac{3}{2} \frac{WL}{AE}$$

to (1),  $V_c = (2+2\sqrt{2}) U_d - 3 \frac{WL}{AE}$

~~$$\therefore V_b = (-\frac{17}{2} + 12\sqrt{2}) \frac{WL}{AE}, V_c = (-4 + 5\sqrt{2}) \frac{WL}{AE}, U_c = \frac{9}{2} \frac{WL}{AE},$$~~

~~$$V_d = (-\frac{3}{2} + 5\sqrt{2}) \frac{WL}{AE}, U_d = \frac{3}{2} \frac{WL}{AE}, V_0 = 3\sqrt{2} \frac{WL}{AE},$$~~

~~$$U_0 = \frac{1}{2} \frac{WL}{AE}$$~~

$$\therefore \delta(1) = V_c = (-2-\sqrt{2}) \frac{WL}{AE}$$

$$\therefore V_b = -\frac{1}{2} \frac{WL}{AE}, V_c = (-2-\sqrt{2}) \frac{WL}{AE}, U_c = \frac{3}{2} \frac{WL}{AE}, V_d = (-\frac{3}{2} + \sqrt{2}) \frac{WL}{AE}$$

$$U_d = -\frac{3}{2} \frac{WL}{AE}, V_e = (-6-3\sqrt{2}) \frac{WL}{AE}, U_e = -\frac{5}{2} \frac{WL}{AE}$$

$$\therefore T_1 = \frac{3}{2} W, T_2 = \sqrt{2} W, T_3 = -W, T_4 = -\frac{1}{2} W,$$

$$T_5 = -\frac{3}{2} W, T_6 = \frac{1}{2} W, T_7 = \frac{\sqrt{2}}{2} W, T_8 = -W - \frac{\sqrt{2}}{2} W$$



3/23/2023 Pin-jointed During Lab

|              | width (mm) | height (mm) |
|--------------|------------|-------------|
|              |            |             |
| Mesurement 1 |            |             |
| Mesurement 2 |            |             |
| Mesurement 3 |            |             |
| Average      |            |             |
| Area         |            |             |



3/23/2023 Pin-jointed During Lab

|           | A<br>(m <sup>2</sup> ) | E<br>(Pa) | L<br>(m) | Load<br>(N) | Deflection<br>(mm) |   |   | Tension 1<br>(N) |   |   | Tension 2<br>(N) |   |   | Tension 3<br>(N) |   |   | Tension 4<br>(N) |   |   | Tension 5<br>(N) |   |   | Tension 6<br>(N) |   |   | Tension 7<br>(N) |   |   |
|-----------|------------------------|-----------|----------|-------------|--------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|
|           |                        |           |          |             | 1                  | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 |
| Loading   |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |
|           |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |
|           |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |
| Unloading |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |
|           |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |
|           |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |

Statically determinate



3/23/2023 Pin-jointed During Lab

|           | A<br>(m <sup>2</sup> ) | E<br>(Pa) | L<br>(m) | Load<br>(N) | Deflection<br>(mm) |   |   | Tension 1<br>(N) |   |   | Tension 2<br>(N) |   |   | Tension 3<br>(N) |   |   | Tension 4<br>(N) |   |   | Tension 5<br>(N) |   |   | Tension 6<br>(N) |   |   | Tension 7<br>(N) |   |   | Tension 8<br>(N) |   |   |
|-----------|------------------------|-----------|----------|-------------|--------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|
|           |                        |           |          |             | 1                  | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 |
| Loading   |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |
|           |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |
|           |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |
| Unloading |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |
|           |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |
|           |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |

Statically Indeterminate



3 / 23 / 2023 Pre-jointed Post-Lab

|           |  | A<br>(m <sup>2</sup> ) | E<br>(Pa) | L<br>(m) | Load<br>(N) | Deflection<br>(mm) |   |   | Tension 1<br>(N) |   |   | Tension 2<br>(N) |   |   | Tension 3<br>(N) |   |   | Tension 4<br>(N) |   |   | Tension 5<br>(N) |   |   | Tension 6<br>(N) |   |   | Tension 7<br>(N) |   |   |  |  |
|-----------|--|------------------------|-----------|----------|-------------|--------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|--|--|
|           |  |                        |           |          |             | 1                  | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 |  |  |
|           |  |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |  |  |
| Loading   |  |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |  |  |
|           |  |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |  |  |
|           |  |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |  |  |
| Unloading |  |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |  |  |
|           |  |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |  |  |
|           |  |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |  |  |

Statically determinate



3/24/2023 Pin-jointed Post-Lab

|           | A<br>(m <sup>2</sup> ) | E<br>(Pa) | L<br>(m) | Load<br>(N) | Deflection<br>(mm) |   |   | Tension 1<br>(N) |   |   | Tension 2<br>(N) |   |   | Tension 3<br>(N) |   |   | Tension 4<br>(N) |   |   | Tension 5<br>(N) |   |   | Tension 6<br>(N) |   |   | Tension 7<br>(N) |   |   | Tension 8<br>(N) |   |   |
|-----------|------------------------|-----------|----------|-------------|--------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|------------------|---|---|
|           |                        |           |          |             | 1                  | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 | 1                | 2 | 3 |
| Loading   |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |
|           |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |
|           |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |
| Unloading |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |
|           |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |
|           |                        |           |          |             |                    |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |                  |   |   |

Statically Indeterminate



3/24/2023 Pin-jointed Post-Lab

|            | A<br>(m <sup>2</sup> ) | E<br>(Pa) | L<br>(m)  | Load<br>(N) | Deflection |       | Tension 1 |       | Tension 2 |       | Tension 3 |       | Tension 4 |       | Tension 5 |       | Tension 6 |       | Tension 7 |       |
|------------|------------------------|-----------|-----------|-------------|------------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|
|            |                        |           |           |             | (mm)       | error | (N)       | error | (N)       | error | (N)       | error | (N)       | error | (N)       | error | (N)       | error | (N)       | error |
| Experiment |                        |           | Loading   |             |            |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            |                        |           |           |             |            |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
| Theory     |                        |           | Unloading |             |            |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            |                        |           |           |             |            |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            |                        |           | Loading   |             |            |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            |                        |           |           |             |            |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |

Statically Determinate



3/24/2023 Pin-jointed Post-Lab

|            | A<br>(m <sup>2</sup> ) | E<br>(Pa) | L<br>(m) | Load<br>(N) | Deflection<br>(mm) |      | Tension 1 |       | Tension 2 |       | Tension 3 |       | Tension 4 |       | Tension 5 |       | Tension 6 |       | Tension 7 |       | Tension 8 |       |
|------------|------------------------|-----------|----------|-------------|--------------------|------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|
|            |                        |           |          |             | error              | (mm) | (N)       | error | (N)       | error | (N)       | error | (N)       | error | (N)       | error | (N)       | error | (N)       | error | (N)       | error |
| Experiment | Loading                |           |          |             |                    |      |           |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            |                        |           |          |             |                    |      |           |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            |                        |           |          |             |                    |      |           |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            |                        |           |          |             |                    |      |           |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            |                        |           |          |             |                    |      |           |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            | Unloading              |           |          |             |                    |      |           |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            |                        |           |          |             |                    |      |           |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            |                        |           |          |             |                    |      |           |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            |                        |           |          |             |                    |      |           |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            |                        |           |          |             |                    |      |           |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
| Theory     | Loading                |           |          |             |                    |      |           |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            |                        |           |          |             |                    |      |           |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            |                        |           |          |             |                    |      |           |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            |                        |           |          |             |                    |      |           |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            |                        |           |          |             |                    |      |           |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            | Unloading              |           |          |             |                    |      |           |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            |                        |           |          |             |                    |      |           |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            |                        |           |          |             |                    |      |           |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            |                        |           |          |             |                    |      |           |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |
|            |                        |           |          |             |                    |      |           |       |           |       |           |       |           |       |           |       |           |       |           |       |           |       |

Statically Indeterminate



3/24/2023 Pin-jointed Post-Lab

Statically Determinate



3/24/2023 Pin-jointed Post-Lab



3/24/2023 Pin-jointed Post-Lab

Statically Indeterminate

Deflection (mm)

Tension 4 (N)

Tension 8 (N)



3/24/2023 Pin-jointed Post-Lab

